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SARDAR PATEL UNIVERSITY

BCA Semester-2 External Examination – April-2025 (New Course) [NEP]

US02IDBCA05 || Digital Electronics

Date : 16-04 -2025

Time : 02.00 p.m. – 03.30 p.m.

Total : 25 Marks

Q-1 Select the appropriate option for the following questions: [05]

- 1) A _____ is a combinational circuit that converts binary information from the “n” coded inputs to “2ⁿ” unique outputs.

A. Half Adder B. Decoder
C. Encoder D. Comparator

- 2) De Morgan’s first theorem says that a NOR gate is equivalent to a _____

A. Bubbled OR B. Bubbled NOR
C. Bubbled AND D. Bubbled NAND

- 3) The NOR gate has two or more input signals. If all inputs are _____, the output is high.

A. High B. Both A and B
C. Low D. None of these

- 4) What is the one disadvantage of an RS Flip-Flop has _____

A. no Enable input B. a RACE condition
C. no clock input D. only single output

- 5) A register that is used to store binary information is called _____

A. Binary Register B. Data Register
C. Shift Register D. None of these

Q-2 Define logic gate. Draw electronic symbol and explain AND, OR, NOT and XOR gates with truth table. [10]

OR

Q-2 Explain Half adder and Full Adder with circuit and truth table. [10]

Q-3 Explain D Flip-Flop with diagram. [10]

OR

Q-3 Explain Left-Shift and Right-Shift register with diagram. [10]

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